

Software Design Document (SDD)

GTIDS Online Agreement Management & Digital Signing Portal

Organization: Gramtarang Inclusive Development Services (GTIDS)

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1. Design Objectives

- Keep the GTIDS agreement platform independent of any commercial agreement-management SaaS.
- Use the third-party provider only as the Aadhaar OTP/eSign integration boundary.
- Support physical 100 stamp paper now and allow e-Stamp integration later.
- Guarantee sequential signing and approval: Agent → Employee → MD.
- Preserve document integrity, signatures and audit evidence.
- Allow the signing provider to be replaced through an adapter interface.

2. High-Level Architecture

Internet → WAF/Reverse Proxy → Web UI/API →

Agreement/Template/Stamp/Identity/eSign/Workflow/Document/Audit modules → PostgreSQL/Redis/Object Storage.

Technology/Design

Layer	Technology/Design
Web UI	Next.js/React + TypeScript
API	Node.js + NestJS + TypeScript
Database	PostgreSQL
Cache/Queue	Redis + BullMQ
Document Storage	MinIO or S3-compatible private object storage
PDF Engine	Playwright/Chromium or equivalent internally hosted PDF renderer
Reverse Proxy	Nginx
External Integration	Aadhaar OTP/eSign provider; optional DigiLocker
Monitoring	Prometheus/Grafana or equivalent internally hosted monitoring

3. Core Modules

Module	Design Responsibility
Authentication & RBAC	Users, sessions, roles, permissions and administrative security.
Agreement Management	Agreement creation, status, metadata, parties and lifecycle.
Template Management	Template versions, variables, approval and retirement.
Stamp Management	Physical stamp inventory, scan storage, allocation and usage tracking.
Identity Module	Aadhaar OTP transaction orchestration and verification result handling.

Module	Design Responsibility
DigiLocker Module	Optional consent-based document retrieval/verification.
eSign Module	Signing transaction orchestration and provider adapter.
Document Module	PDF generation, versioning, hashing, storage and retrieval.
Workflow Module	Agent signature → Employee approval → MD signature state machine.
Audit Module	Immutable/append-oriented event logging.
Notification Module	Completion email to Agent, Employee and MD; operational alerts.
Verification Module	QR/public verification endpoint.
Reporting Module	Operational and management reports.

4. Detailed Workflow Design

Step	Design
1. Initiation	Agent creates agreement and selects agreement type/template.
2. Data Validation	System validates required fields and references.
3. Stamp	Physical 100 stamp is uploaded/allocated or future e-Stamp service is invoked.
4. Generation	Document engine merges template + data + stamp and creates Version 1.
5. Agent Review	Agent previews the complete document.
6. Agent Sign	System creates eSign transaction; third party handles Aadhaar OTP/eSign.
7. Agent Completion	Signed document is stored; state becomes AGENT_SIGNED.
8. Employee Approval	Employee reviews document and approves/rejects.
9. MD Queue	On approval, state becomes PENDING_MD_SIGNATURE.
10. MD Sign	MD reviews and signs through the configured eSign provider.
11. Finalization	System verifies MD signing, stores final document and sets COMPLETED.
12. Notification	System emails final agreement to Agent, Employee and MD.
13. Verification	System publishes a non-sensitive verification record and QR.

5. State Machine

DRAFT → READY_FOR_AGENT_SIGNATURE → AGENT_SIGNING → AGENT_SIGNED → PENDING_EMPLOYEE_APPROVAL → EMPLOYEE_APPROVING → EMPLOYEE_APPROVED → PENDING_MD_SIGNATURE → MD_SIGNING → COMPLETED. Failure/exception transitions: REJECTED, CANCELLED, EXPIRED, SIGNATURE_FAILED.

6. Component Design

Recommended project structure:

- src/auth/
- src/users/
- src/roles/
- src/agreements/
- src/templates/
- src/stamps/

- src/identity/
- src/digilocker/
- src/esign/
- src/documents/
- src/workflow/
- src/notifications/
- src/audit/
- src/verification/
- src/reports/

7. Database Design

Table	Purpose
users	Internal user accounts and profile information.
roles / permissions / user_roles	RBAC.
agreement_types	Types of agreements.
agreement_templates	Logical template records.
agreement_template_versions	Versioned template content and approval metadata.
agreements	Core agreement record and lifecycle state.
agreement_parties	Agent, Employee, MD and other parties.
stamp_papers	Physical stamp inventory.
stamp_allocations	Mapping of a stamp to an agreement.
identity_verifications	Aadhaar verification transactions and outcomes.
digilocker_transactions	Optional DigiLocker consent/transaction metadata.
esign_transactions	Provider signing transactions.
signature_events	Per-signer signature lifecycle events.
agreement_versions	Every generated document version and hash.
agreement_documents	Object storage references and document metadata.
audit_logs	Security and business event history.
notifications	Email/notification records and delivery status.

8. Key Database Fields

Table	Important Fields
agreements	id, agreement_number, agreement_type_id, template_version_id, status, stamp_type, current_version, created_by, created_at, completed_at
agreement_parties	id, agreement_id, party_type, name, email, mobile, identity_reference, signing_order, status
stamp_papers	id, stamp_number, denomination, state, issue_date, vendor, file_key, document_hash, status
esign_transactions	id, agreement_id, party_id, provider, provider_transaction_id, document_hash, status, initiated_at, completed_at
agreement_versions	id, agreement_id, version_no, document_hash, file_key, created_at
audit_logs	id, agreement_id, actor_id, event_type, event_data, ip_address, user_agent, timestamp

9. API Design

Endpoint	Purpose
POST /api/v1/agreements	Create agreement.
GET /api/v1/agreements/:id	Retrieve agreement.
POST /api/v1/agreements/:id/generate	Generate agreement PDF.
POST /api/v1/stamps	Create/register physical stamp.
POST /api/v1/stamps/:id/allocate	Allocate stamp to agreement.
POST /api/v1/identity/initiate	Start identity verification.
GET /api/v1/identity/:id/status	Check verification status.
POST /api/v1/esign/initiate	Start Agent/MD signing.
GET /api/v1/esign/:id/status	Check provider transaction.
POST /api/v1/esign/callback	Receive provider callback.
POST /api/v1/agreements/:id/employee-approve	Employee approval.
POST /api/v1/agreements/:id/reject	Reject with reason.
GET /api/v1/agreements/:id/audit	Retrieve audit trail.
GET /api/v1/agreements/:id/document	Retrieve authorized document.
GET /api/v1/verify/:agreementNumber	Public verification.

10. eSign Provider Adapter

Provider-specific implementation shall be isolated behind an internal interface. The agreement system shall not contain provider-specific logic in the workflow module.

Interface	Purpose
initiateSigning()	Create signing request for signer and document hash.
getStatus()	Retrieve signing transaction status.
handleCallback()	Validate and process provider callback.
getSignedDocument()	Retrieve completed signed document when supported.
validateTransaction()	Validate transaction integrity/status.

11. Document Architecture

Documents shall be stored in private object storage using an agreement-centric path such as agreements/{year}/{month}/{agreement-number}/.

- Original stamp scan
- Generated unsigned PDF
- Agent-signed document
- Employee approval record
- MD-signed/final PDF
- Audit certificate/export where required

Every significant document version shall have a SHA-256 hash.

12. Email Architecture

After finalization, the Notification Module shall create three recipient records: Agent, Employee and MD. The same final agreement reference and final signed document/link shall be sent to all three.

Event	Recipients
Agent signed	Employee / MD task notification as configured; not final completion.
Employee approved	MD task notification.
MD signed / Agreement completed	Agent + Employee + MD.
Agreement rejected	Relevant prior actor(s) and initiator.

13. Security Architecture

- TLS for all external and internal HTTP traffic where applicable.
- RBAC with least privilege.
- MFA for privileged users.
- Encrypted database/storage backups.
- Private object storage; no public document buckets.
- Short-lived, authorization-checked download URLs.
- No storage of Aadhaar OTP.
- Provider transaction IDs and permitted identity references only.
- API rate limiting and request validation.
- Audit logging for all signature and approval actions.
- Secrets stored outside source code.
- Regular vulnerability scanning and penetration testing before production.

14. Deployment Architecture

Component	Production Recommendation
Web	Next.js behind Nginx/WAF.
API	2+ application instances behind load balancer when required.
PostgreSQL	Dedicated database server/cluster with backups.
Redis	Dedicated Redis instance with persistence where required.
Object Storage	Private MinIO/S3-compatible storage with versioning.
Worker	Separate worker process for PDF generation, email and background jobs.
Monitoring	Centralized logs, metrics and alerts.
Backup	Separate backup location and periodic restore tests.

15. Background Jobs

- PDF generation
- Email sending
- Provider status reconciliation
- Reminder notifications

- Audit export
- Backup verification
- Document integrity checks

16. Logging and Monitoring

Log Type	Examples
Application	Errors, warnings, request IDs.
Workflow	State transitions and approval events.
Signature	Provider transaction events and callback results.
Security	Login failures, privilege changes, suspicious activity.
Audit	Business/legal significant actions.
Infrastructure	CPU, memory, disk, database, queue and storage health.

17. Testing Strategy

- Unit tests for business rules and state transitions.
- API integration tests.
- PDF generation and document-hash tests.
- Provider sandbox integration tests.
- End-to-end Agent → Employee → MD workflow tests.
- Email delivery tests.
- Role/permission tests.
- Security tests including OWASP Top 10 coverage.
- Load and concurrency testing.
- Backup/restore testing.
- Signature/document integrity testing.

18. Critical Business Rules

Rule	Requirement
BR-001	Only an authorized Agent can initiate/sign an assigned agreement.
BR-002	Employee approval is disabled until Agent signature is successful.
BR-003	MD signing is disabled until Employee approval is successful.
BR-004	Any rejection returns the agreement to a controlled correction/re-initiation state.
BR-005	Completed agreements cannot be edited; corrections require a new version/re-execution policy.
BR-006	A physical stamp cannot be reused for another completed agreement.
BR-007	Final completion occurs only after successful MD signature.
BR-008	Completion email is sent only after finalization.
BR-009	Audit entries cannot be deleted by ordinary application users.
BR-010	Public verification shall expose only non-sensitive information.

19. Recommended Initial Implementation Plan

1. Set up repository, CI/CD, PostgreSQL, Redis and private document storage.
2. Implement authentication, roles and admin controls.
3. Implement agreement types/templates and PDF generation.
4. Implement physical stamp inventory and allocation.
5. Implement Agent → Employee → MD workflow/state machine.
6. Integrate Aadhaar OTP/eSign provider through adapter.
7. Implement email notification to all three parties after completion.
8. Implement audit trail and document hashing.
9. Implement QR verification.
10. Complete security, load, recovery and end-to-end testing.

20. Key Design Decision

The GTIDS platform remains the system of record. The third-party Aadhaar/eSign service is an external identity/signature dependency only. Provider-specific APIs must be isolated in an adapter so that the GTIDS platform does not become coupled to a particular vendor.